WATER RESOURCES, WELLS & PUMPS · NOTE WR07

Groundwater Regions, Development, Legislation and Conjunctive Use

Course topics: IDE6.2 L3, IDE-506 T1, IDE-506 T5, IDE-503 T12

Groundwater regions of India

Groundwater is the most important replenishable mineral resource; its occurrence depends on the underground formations and regional rainfall. The Geological Survey of India divides the country into three geo-hydrological units:

Unit	Formations	Extent and yield	Where
Hard rock areas	Archean crystalline and Deccan trap formations	About 120 million ha; low yield; tapped mainly by open wells	Rajasthan, MP, Bihar, Gujarat, Maharashtra, Andhra Pradesh, Odisha, Karnataka, Tamil Nadu, Kerala
Semi-consolidated formations	Sandstones, limestones, conglomerates	About 5 million ha; moderate yield	Rajasthan, Gujarat, Tamil Nadu
Alluvial formations	Sand, gravel, boulders, laterites, soils, clays	Kashmir valley, Indus basin, Ganga-Brahmaputra-Meghna basin, coastal tracts; substantial yield	Punjab, Haryana, UP, Bihar, West Bengal, Assam

Assessment and recharge

- Scientific assessment began with the Five-Year Plans after 1951; the Geological Survey of India mapped the formations systematically.
- An early rough assessment by the Central Ground Water Board put the annual utilisable resource at 423 km³/year, with present utilisation of 100 km³/year (≈ 24%). (CGWB now assesses the resource dynamically — 2023: 407.21 BCM extractable, 59.26% extracted.)
- **Artesian aquifers:** a narrow artesian belt runs below the Himalayas from Himachal Pradesh through Uttar Pradesh to Bihar; others occur in eastern coastal Tamil Nadu. They have large scope for recharge.
- **Recharge** comes mainly from rainfall, supplemented by seepage from canals, field channels, tanks and ponds, influent rivers and deep percolation from irrigated fields.
- **Raise recharge** by soil conservation, terracing, contour bunding, gully plugging and intensive recharge programmes.

Progressive development

Development took the easier, cheaper sources first and is now moving to deeper, costlier ones while domestic, industrial and livestock demand keeps rising. This calls for recharge enhancement and restrictions on indiscriminate pumping.

Legislation

- Potential is limited and region-specific; indiscriminate pumping upsets underground hydrology and depletes the resource.
- State Irrigation Acts do not clearly define who owns groundwater.
- The Government of India circulated a model **Ground Water Control and Regulation Bill in 1970** (revised several times since).
- The **Irrigation Commission (1972)** recommended that States enact laws to regulate groundwater use.
- CGWB prepares a **Ground Water Year Book** every year.

Conjunctive use of surface water and groundwater

Many canal systems — in northern India often only ¼ to ⅓ of what intensive farming needs — give water that is neither enough nor on time. Combined (conjunctive) use of canal water and groundwater fills the gap.

Form	Objective
Use all canal water, supplemented by groundwater	Augment canal supplies
Pump groundwater during low canal supply or canal closure	Combat waterlogging by lowering the water table
Irrigate part of the command only with groundwater	Make poor-quality groundwater usable (by blending)

- **Western Yamuna Canal (Haryana):** tubewells along the canal banks augment the canal supply and lower the water table.
- **Israel and the USA** have used saline groundwater, diluted with good surface water, to a limited extent; there is scope for the same in parts of **Gujarat and Rajasthan**, where groundwater is brackish.
- **Planning:** quantify both resources region by region, study how they interact, and use economic analysis to choose the best mode of use.

Sources: ICAR eCourse 'Groundwater, Wells and Pumps' — Dr M. K. Jha (IIT Kharagpur), reviewed by Dr P. B. S. Sarma (IARI New Delhi). Central Water Commission, Reassessment of Water Availability in India using Space Inputs (2019); Rajya Sabha reply (Dec 2022) on per-capita availability; CGWB Dynamic Ground Water Resources of India (2023 and 2025 assessments); CGWB & Ground Water Department, Govt of Rajasthan, Dynamic Ground Water Resources of Rajasthan 2024; WHO fact sheet 'Diarrhoeal disease' (7 March 2024). Further reading: A. M. Michael and T. P. Ojha, Principles of Agricultural Engineering, Vol. II (Jain Brothers).

Practice MCQs

Q1. Hard rock areas of India cover about

(a) 5 million ha
(b) 50 million ha
(c) 120 million ha
(d) 300 million ha

Q2. Alluvial formations with substantial groundwater yield are found mainly in

(a) Deccan plateau
(b) Indo-Gangetic plains
(c) Western Ghats
(d) Kutch

Q3. The model Ground Water Control and Regulation Bill was first circulated in

(a) 1951 (b) 1970

(c) 1987 (d) 2012

Q4. Tubewells along the Western Yamuna Canal banks are an example of

(a) artificial recharge (b) conjunctive use

(c) drip irrigation (d) sub-surface drainage

Q5. Supply in many northern canal systems is only about ___ of intensive needs

(a) ¼ to ⅓ (b) ½ to ¾

(c) all (d) twice

Answers

Q1 — (c) 120 million ha

Q2 — (b) Indo-Gangetic plains

Q3 — (b) 1970

Q4 — (b) conjunctive use

Q5 — (a) ¼ to ⅓